

STAT 6560
Graphical Methods
Fall 2012

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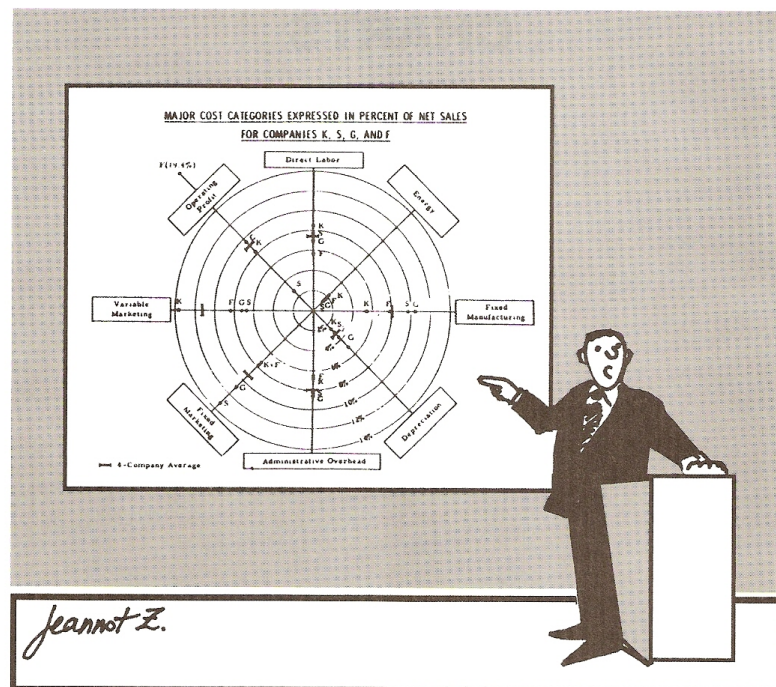
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Acknowledgements

This course uses some of the course materials provided by Dr. Mike Minnotte (formerly USU, now with the University of North Dakota) as held in the Fall 2006 semester. Additional materials have been taken from other Statistical Graphics courses, such as the ones offered by Dr. Di Cook (Iowa State University: <http://www.public.iastate.edu/~dicook/>) and Dr. Dan Carr (George Mason University: <http://mason.gmu.edu/~dcarr/>). We are likely to include parts from additional web sources that will be specified later on.

Thanks are also due to around 20 students and guests who took Stat 6560 with me since the Spring 2009 semester for their valuable comments that helped to improve and correct these lecture notes.

Jürgen Symanzik, August 26, 2012.



"What do you mean, what does it mean?"

Figure 1: Zelazny (2001), p. x, Cartoon.

1 Introduction: A Couple of Good and Bad Examples

(Based on Wainer (1997), Chapter 1 & Tufte (1983), Chapter 2)

1.1 Motivation

Statistical graphics and data visualization are critical elements of modern data analysis and presentation. From initial exploration of a data set to the final presentation of results to the end user, statistical graphics play a vital role in shaping our understanding of our data. Through proper use of graphics, we can make critical discoveries, and communicate them clearly. Conversely, poor use or misuse of graphics can seriously mislead (by accident or design).

As an example, let us look at an example from Fox News, accessible at <http://flowingdata.com/2012/08/06/fox-news-continues-charting-excellence/>. How much higher would the top tax rate be if the Bush tax cuts were to expire at the end of 2012? Really ?!?!?

In this course, we will start with presentation graphics, including discussion of both tools and principles which lead to clear communication and those which serve only to confuse or mislead. We will spend most of the semester in exploratory graphics and data analysis, including data mining. This will be broken down largely by the dimension of the applicable data. One- and two-dimensional datasets require and allow far different methods than those of more than three dimensions. Categorical and regression data call for their own specialized methods.

Even more than most aspects of statistics, graphics and visualization involve art as well as science. In most cases, there are many reasonable approaches. Only an understanding of the options available and the underlying principles will lead to a successful analysis and presentation.

Via graphics, otherwise boring statistical information can become really exciting and entertaining. Hans Rosling (http://en.wikipedia.org/wiki/Hans_Rosling), a Swedish medical doctor and statistician, “sells” statistics extremely well. Enjoy his talk *Debunking Third-World Myths with the Best Stats You’ve Ever Seen*, accessible at <http://www>.

ted.com/index.php/talks/hans_rosling_shows_the_best_stats_you_ve_ever_seen.html. The <http://TED.com> Web page is featured in Time, April 27, 2009, p. 44. In another presentation, Rosling shows an animation of 200 years of health and wealth data of 200 countries in just four minutes: <http://www.flixxy.com/200-countries-200-years-4-minutes.htm>. The underlying software is accessible at <http://www.gapminder.org/> and it is further described in Rosling & Johansson (2009). Google has acquired the software, now called Google Public Data Explorer (<http://www.google.com/publicdata/home>), and they are planning to further extend it. We will revisit this software once we have discussed interactive and dynamic graphics.

1.2 Why Graphics !?!

Why do we need graphics at all. Aren't summary statistics sufficient?

Start R and load the Anscombe (1973) data set. Just type `anscombe` to check whether these data are available — if not, you may have to load the data via:

```
require(stats)
data(anscombe)
```

Then calculate some summary statistics (separately for the four columns of X's and Y's): mean of the X's, mean of the Y's, standard deviation of the X's, standard deviation of the Y's, correlation coefficient, slope and intercept of the regression line, rms error.

http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Anscombe.R

So, the four pairs of X/Y columns basically are identical !?!

Additional details will be provided after class.

1.3 How to Display Data Badly

Wainer (1997), p. 12, states:

“The aim of good data graphics is to display data accurately and clearly.

[...]

Thus, if we wish to display data badly, we have three avenues to follow.

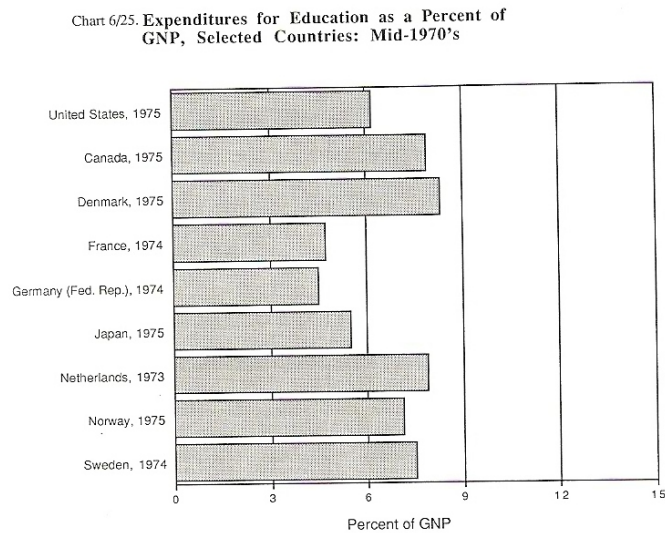
- A. Don't show much data.
- B. Show the data inaccurately.
- C. Obfuscate the data.”[†]

Let us follow these strategies:

[†]Show the data unclearly.

1.3.1 Don't show much data

Rule 1: Show as little data as possible (minimize the data density).



Data Density = 9 numbers / 63 sq. ins. = .14

FIGURE 2. Chart 6/25 from *Social Indicators III* showing expenditures for education for nine countries as a function of GNP.

Figure 2: Wainer (1997), p. 13, Figure 2.

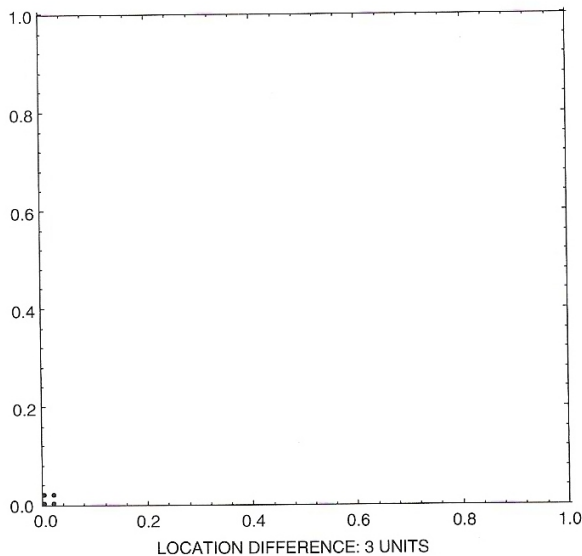


FIGURE 3. A graph of obviously low data density.

Figure 3: Wainer (1997), p. 13, Figure 3.

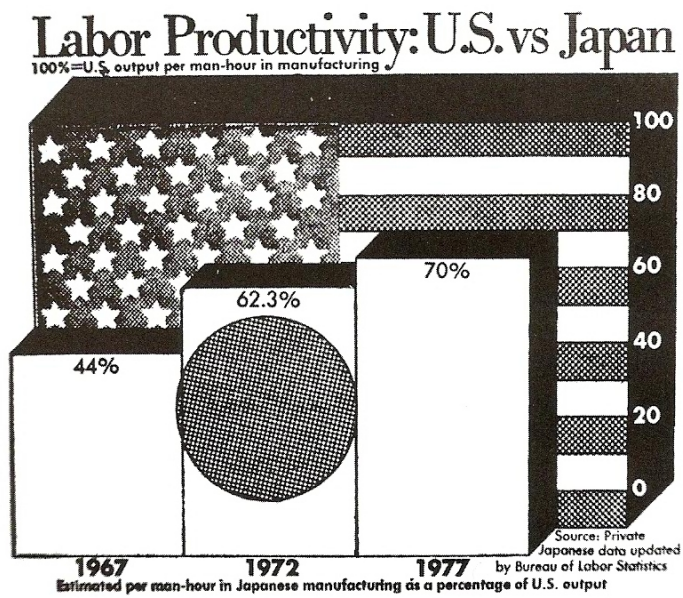


FIGURE 6. A graph with low data density filled in with chartjunk from the *Washington Post*, 1978.

Figure 4: Wainer (1997), p. 16, Figure 6.

Rule 2: Hide what data you do show (minimize the data/ink ratio).

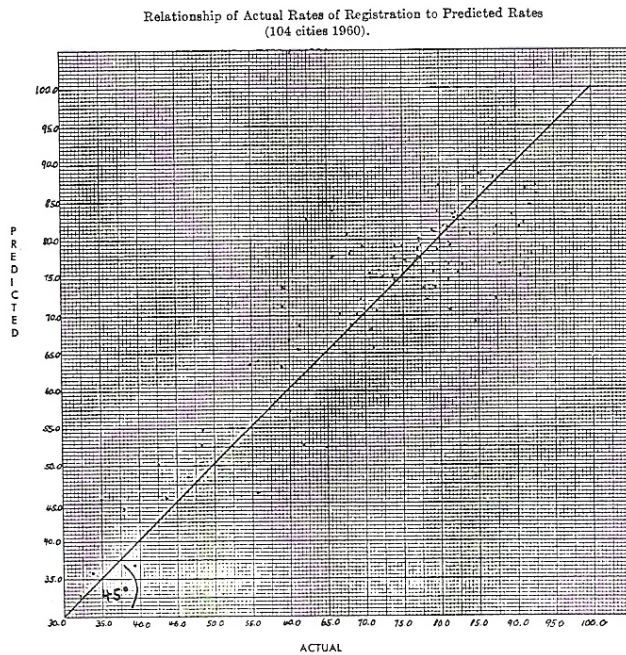


FIGURE 8. Hiding the data in the grid.

Figure 5: Wainer (1997), p. 17, Figure 8: Hiding the data in the grid.

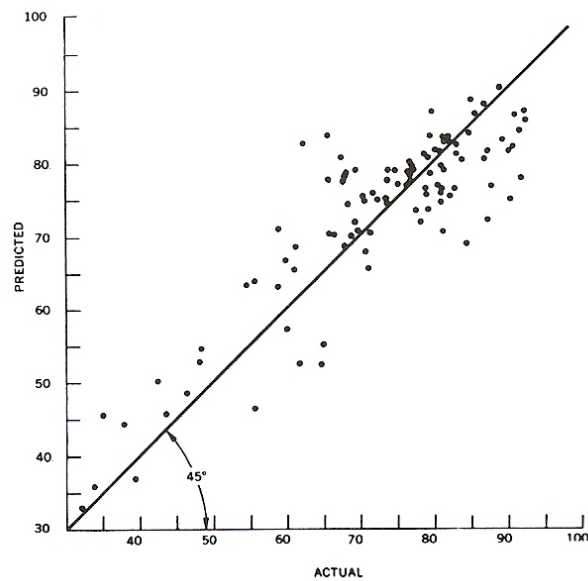
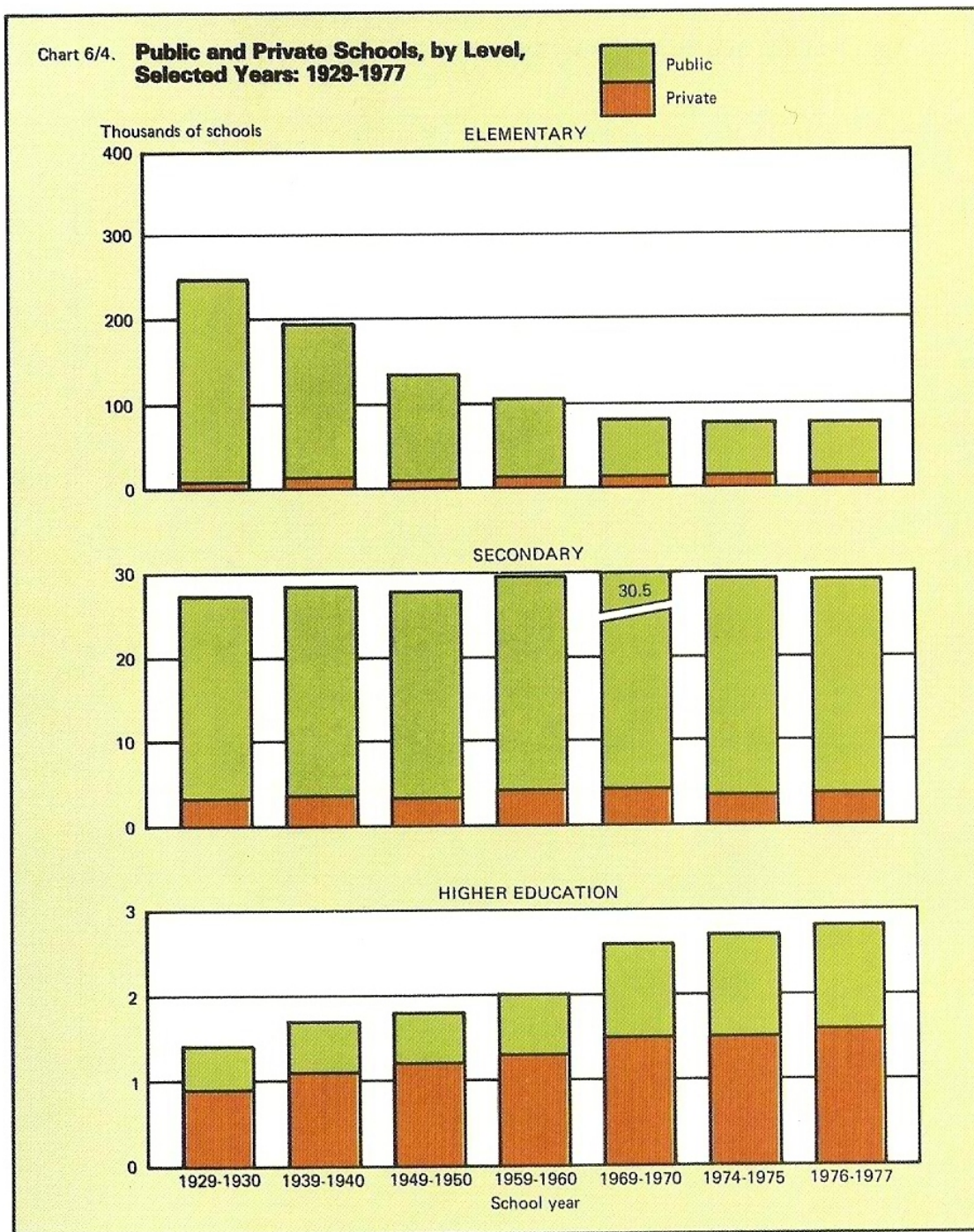


FIGURE 10. A redone example of the data from figure 8.

Figure 6: Wainer (1997), p. 18, Figure 10: Wainer (1997), p. 17, Figure 8, improved.



CHAPTER 1, FIGURE 11. Hiding the data in the scale.

Figure 7: Wainer (1997), p. 20A, Figure 11: Hiding the data in the scale.

FIGURE 12. Expanding the scale and showing the data for the number of private elementary schools from figure 11.

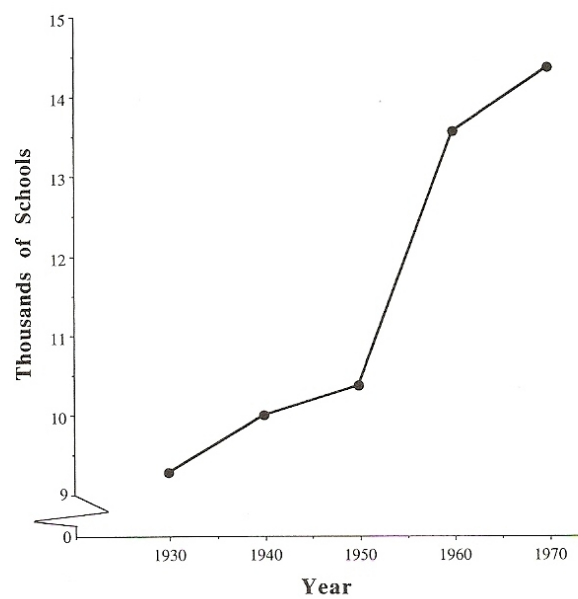


Figure 8: Wainer (1997), p. 20, Figure 12: Wainer (1997), p. 20A, Figure 11, improved.

1.3.2 Show the data inaccurately

Rule 3: Ignore the visual metaphor altogether.

FIGURE 13. Ignoring the visual metaphor by letting a longer bar segment represent a smaller amount of coal (from the *New York Times*, 1978).

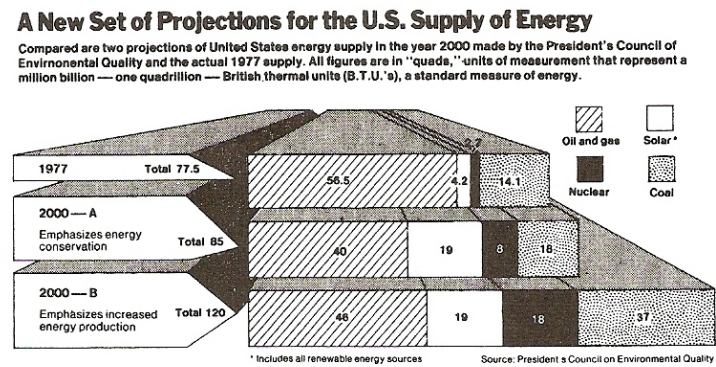


Figure 9: Wainer (1997), p. 20, Figure 13.

U.S. trade with China and Taiwan

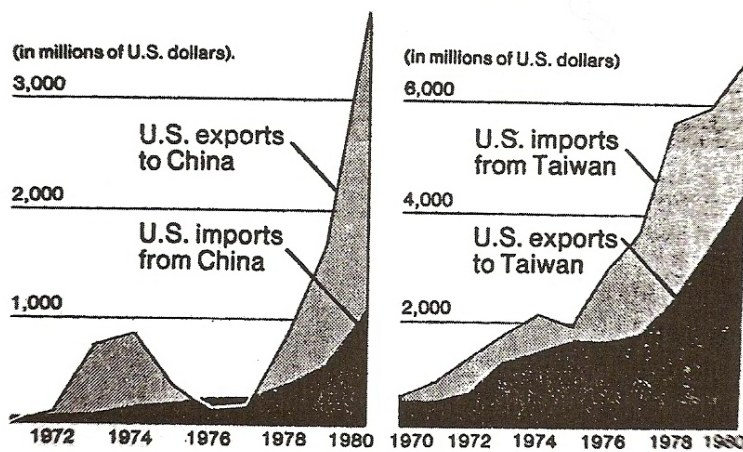


FIGURE 14. Reversing the metaphor in mid-graph while changing scales on both axes (from the *New York Times*, June 14, 1981).

Figure 10: Wainer (1997), p. 21, Figure 14.

FIGURE 15. Figure 14 redone with a consistent scale and visual metaphor.

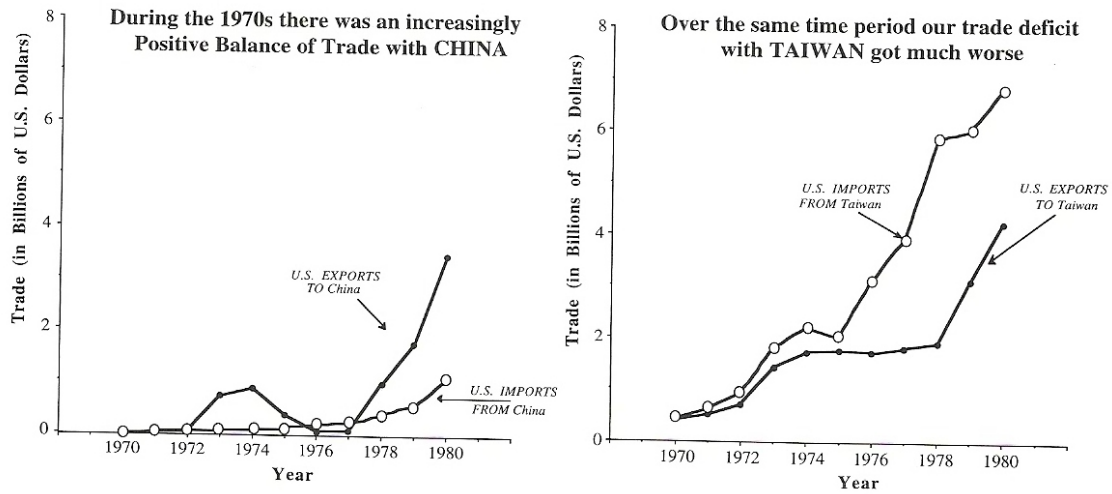


Figure 11: Wainer (1997), p. 21, Figure 15: Wainer (1997), p. 21, Figure 14, improved.

Rule 4: Only order matters.

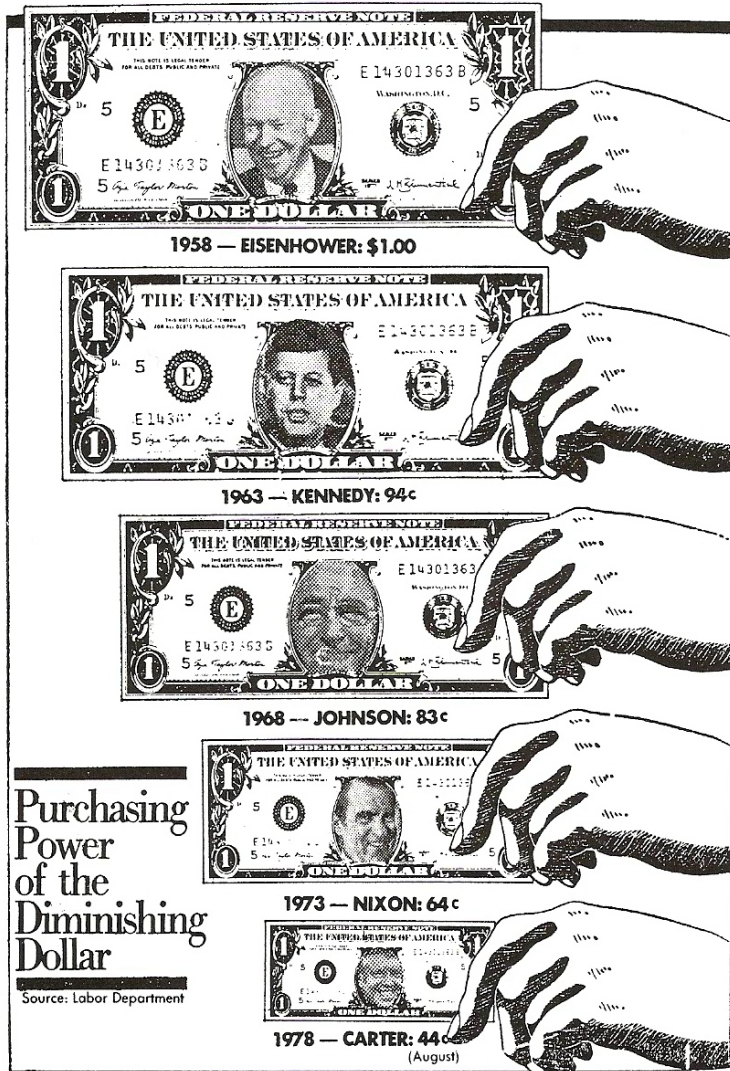


FIGURE 17. An example of how to goose up the effect by squaring the eyeball.

Figure 12: Wainer (1997), p. 23, Figure 17.

Additional details will be provided after class.

Figure 13: Wainer (1997), p. 24, Figure 18: Wainer (1997), p. 23, Figure 17, improved.

FIGURE 19. Cubing the visual effect and choosing the origin to yield a near record lie factor of over 131,000% (from the *Washington Post*).

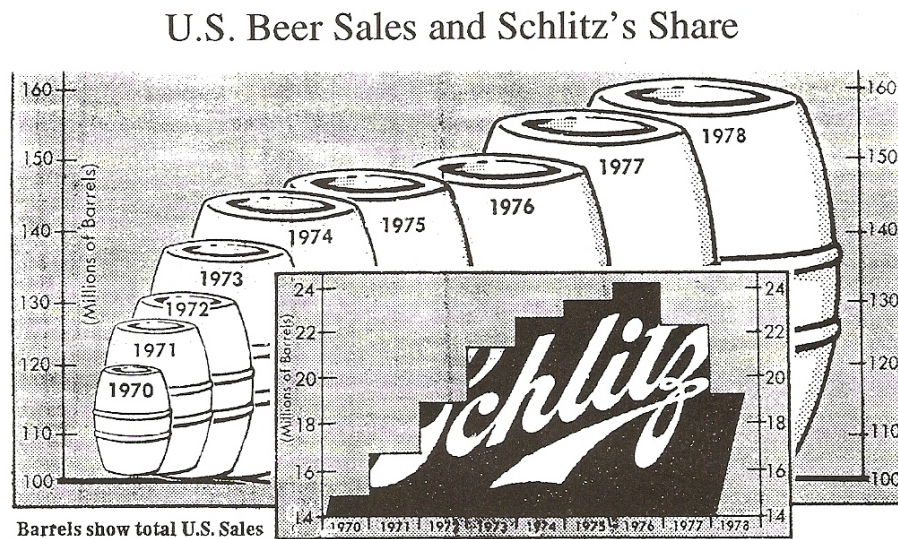


Figure 14: Wainer (1997), p. 24, Figure 19.



FIGURE 20. Data from figure 19 redone without tricks (from Wainer, 1980).

Figure 15: Wainer (1997), p. 25, Figure 20: Wainer (1997), p. 24, Figure 19, improved.

Rule 5: Graph data out of context.

FIGURE 21. Hiding the effect by the careful choice of scale and origin (from the *Washington Post*).

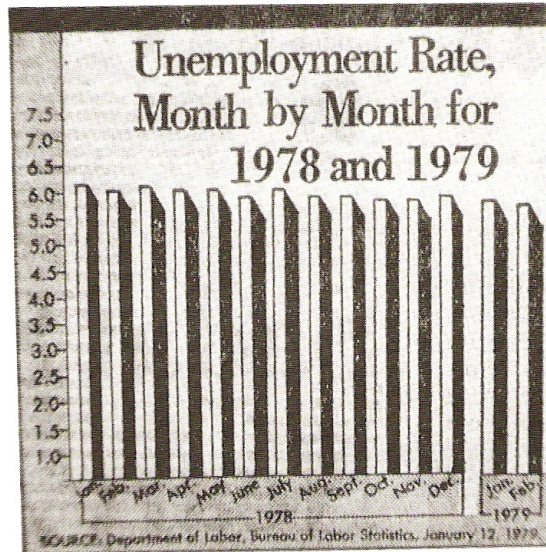


Figure 16: Wainer (1997), p. 26, Figure 21.

FIGURE 22. Regraph of data from figure 21 with expanded scale, different starting point, and previous year's average added for context (from Wainer, 1980).



Figure 17: Wainer (1997), p. 26, Figure 22: Wainer (1997), p. 26, Figure 21, improved.

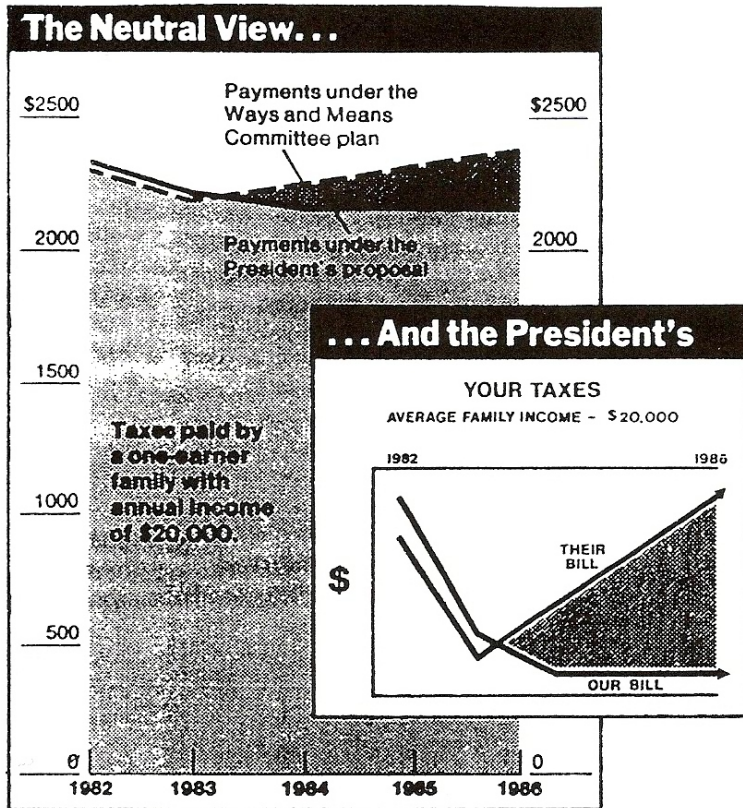


FIGURE 23. *New York Times* graphs showing how lack of context changes our perceptions about alternative tax bills.

Figure 18: Wainer (1997), p. 27, Figure 23.

1.3.3 Obfuscate the data

Rule 6: Change scales in mid-axis.

FIGURE 24. Changing the scale in mid-axis to make large differences seem small (from the *New York Post*, May 12, 1981).

The soaraway Post — the daily paper New Yorkers trust

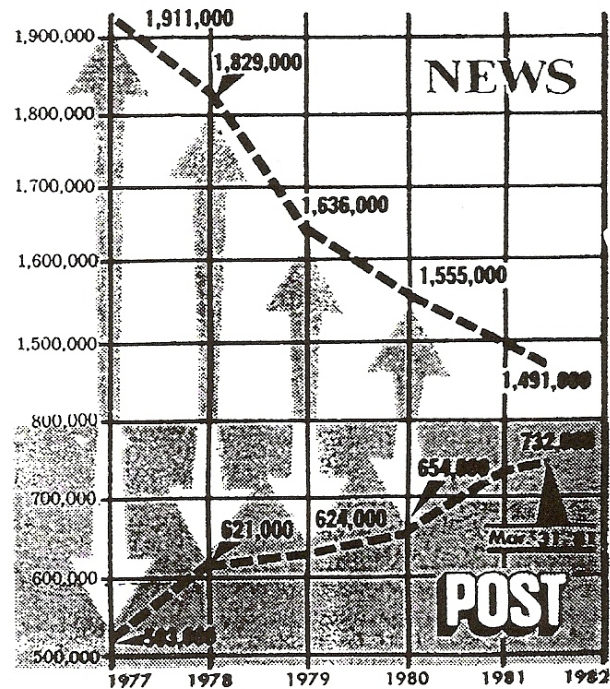


Figure 19: Wainer (1997), p. 28, Figure 24.

FIGURE 25. Changing scale in mid-axis to make exponential growth linear (from the *Washington Post*, Jan. 11, 1979, in an article titled "Pay, Practices of Doctors on Examining Table" by Victor Cohn and Peter Milius).

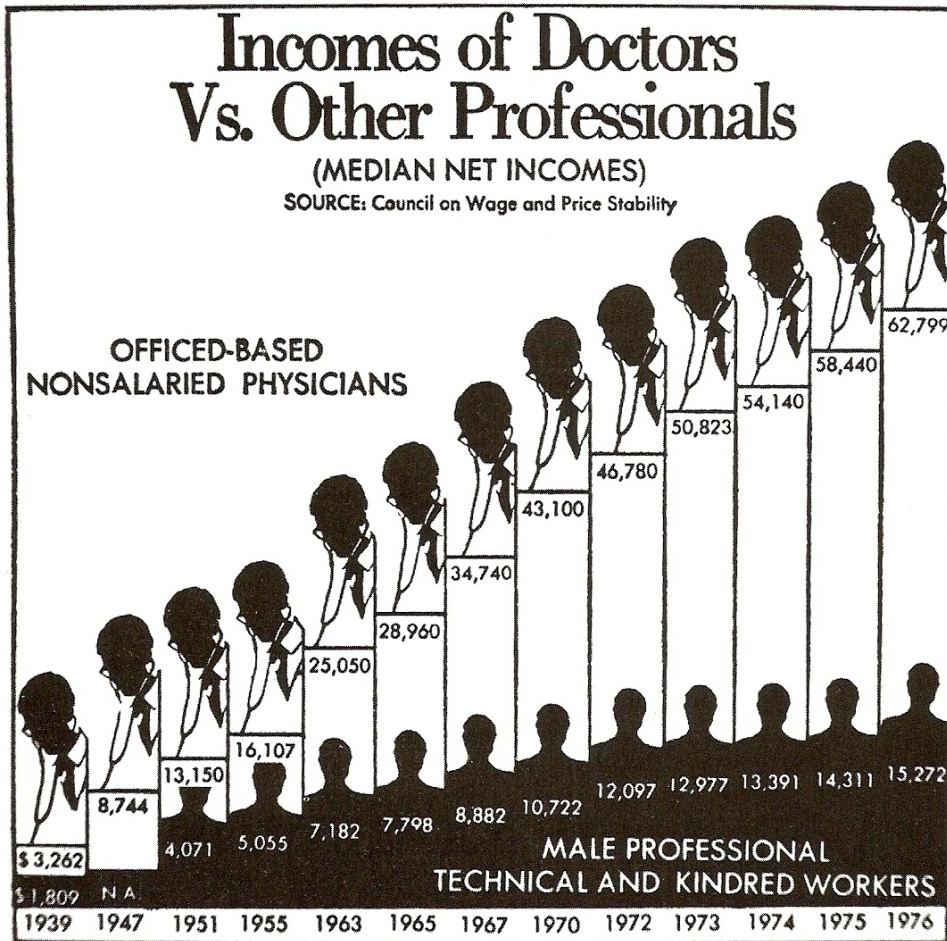


Figure 20: Wainer (1997), p. 29, Figure 25.

FIGURE 26. Data from figure 25 redone with a linear scale (from Wainer, 1980).

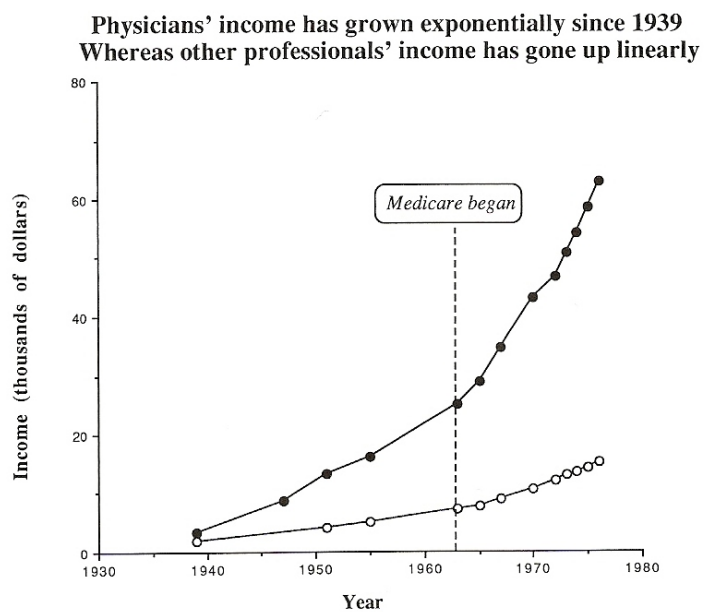
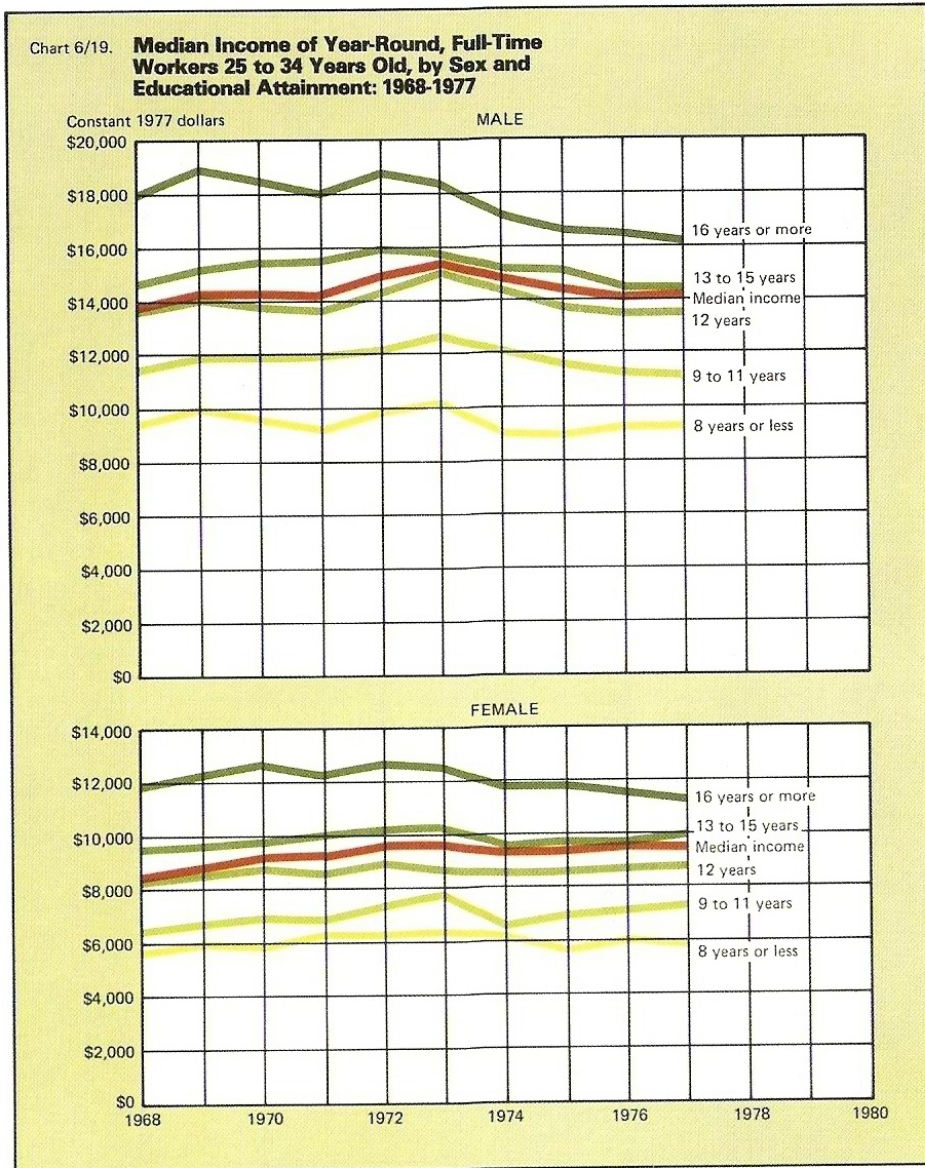


Figure 21: Wainer (1997), p. 30, Figure 26: Wainer (1997), p. 29, Figure 25, improved.

Rule 7: Emphasize the trivial (ignore the important).



CHAPTER 1, FIGURE 27. Emphasizing the trivial: Hiding the main effect of sex differences in income through the vertical placement of plots.

Figure 22: Wainer (1997), p. 20A, Figure 27.

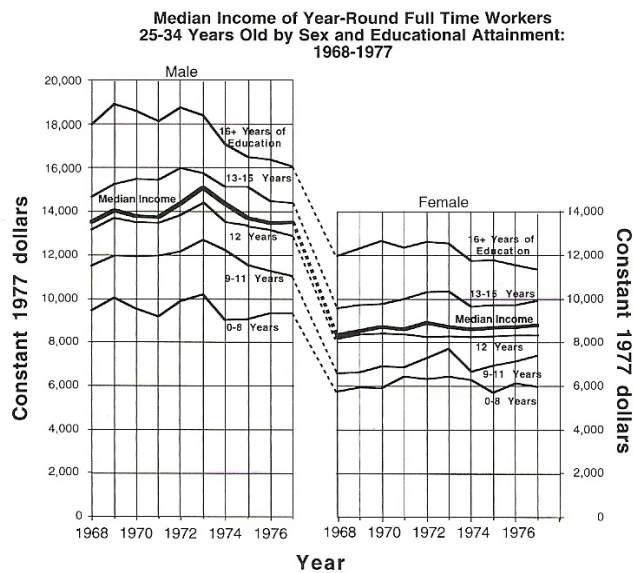


FIGURE 28. Figure 27 redone with the two plots horizontally opposed, showing the size of sex differences more clearly.

Figure 23: Wainer (1997), p. 31, Figure 28: Wainer (1997), p. 20A, Figure 27, improved.

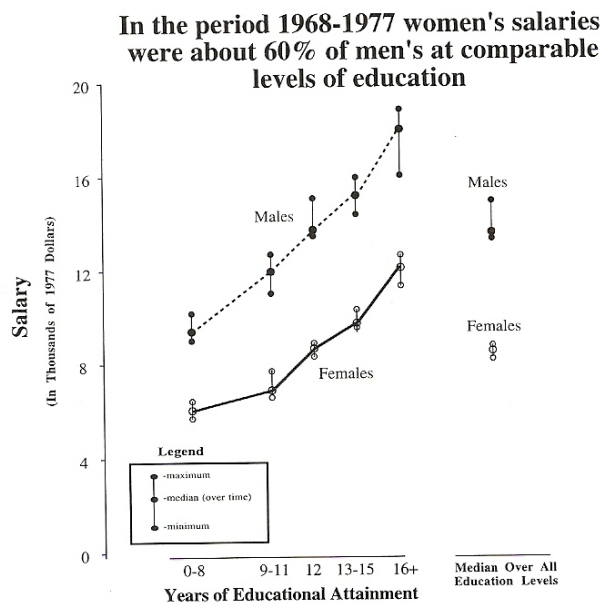
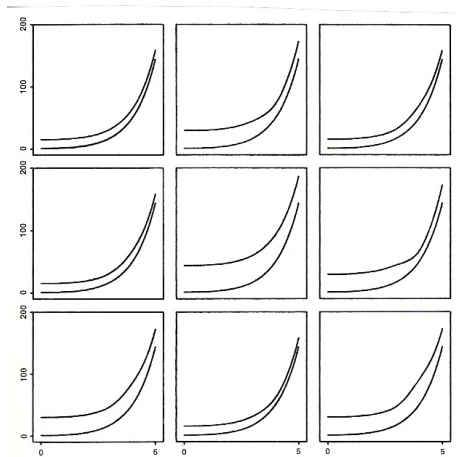


FIGURE 29. Figure 28 redone with the large effects of sex and education emphasized and the small-time trend suppressed.

Figure 24: Wainer (1997), p. 32, Figure 29: Wainer (1997), p. 31, Figure 28, further improved.

Rule 8: Jiggle the baseline.

FIGURE 30. A graphical experiment (from Cleveland and McGill, 1984). Without looking at the corresponding right panel, try to determine the difference between the two curves in the left panel.



Sorry, these plots
got scrambled...

Figure 25: Wainer (1997), p. 33, Figure 30.

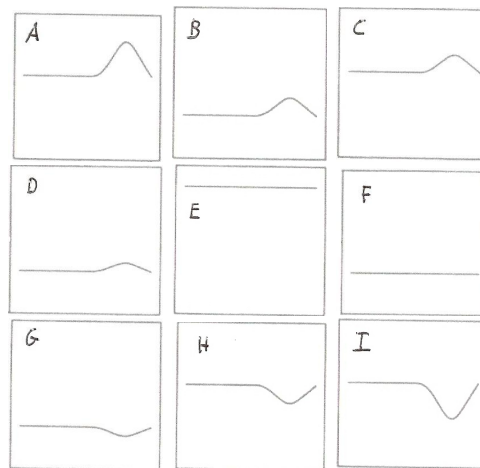
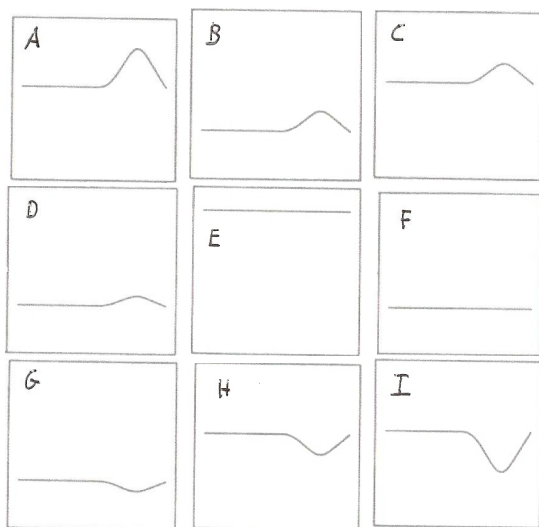
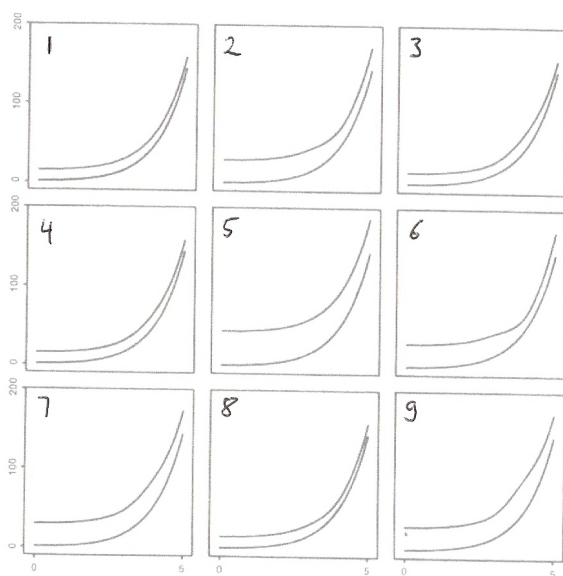


Figure 26: Wainer (1997), p. 33, Figure 30: Scrambled differences. The horizontal axis covers the interval 0 to 5, the vertical axis covers the interval 0 to 50.

Worksheet

Your Name: _____



Task: Match each original (labeled 1 to 9) with the plot (labeled A to I) that shows the difference between upper and lower line in the original plot.

Answer:

1: _____ 2: _____ 3: _____
 4: _____ 5: _____ 6: _____
 7: _____ 8: _____ 9: _____

FIGURE 31. William Playfair's eighteenth-century graph of England's imports and exports with the East Indies (from Cleveland and McGill, 1984).

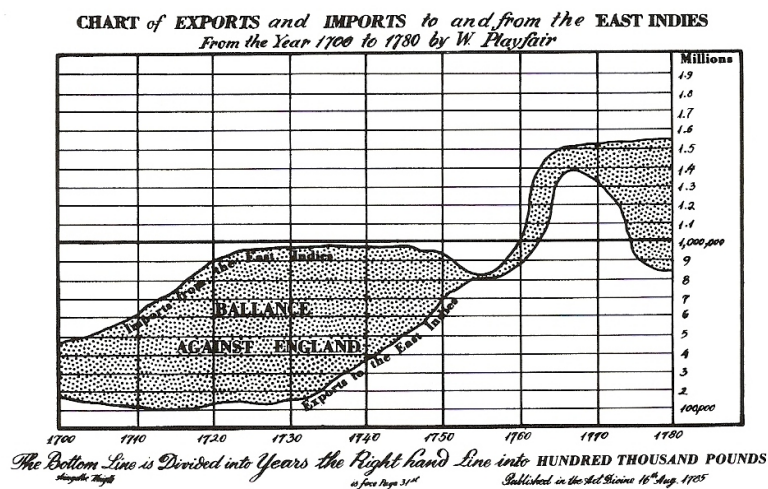


Figure 27: Wainer (1997), p. 34, Figure 31: One of William Playfair's few mistakes.

FIGURE 32. A graph of the difference between West Indies imports and exports showing explicitly the previously invisible jump in the 1760s (from Cleveland and McGill, 1984).

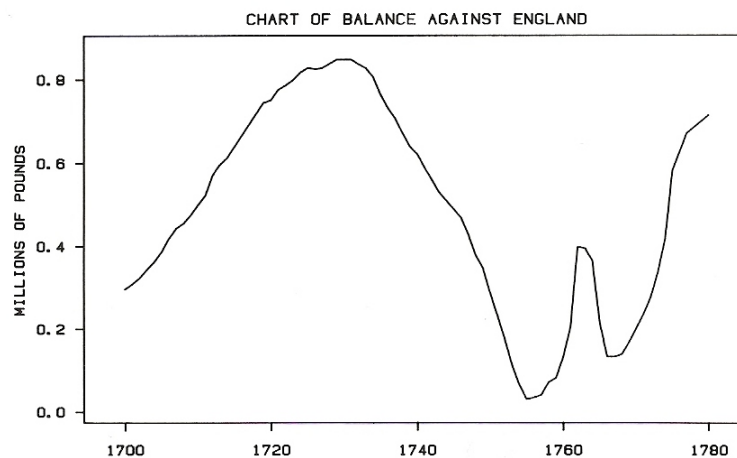


Figure 28: Wainer (1997), p. 34, Figure 32: Wainer (1997), p. 34, Figure 31, improved.

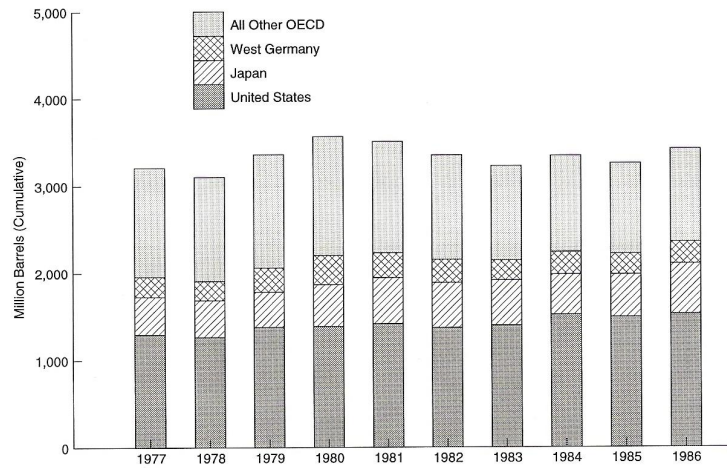


FIGURE 33. From the U.S. Department of Energy's *Annual Energy Review, 1986*, showing the changes in primary stocks of petroleum in OECD countries.

Figure 29: Wainer (1997), p. 35, Figure 33.

OECD PETROLEUM STOCKS HAVE STABILIZED
But Not All Countries Are Pulling Their Own Weight

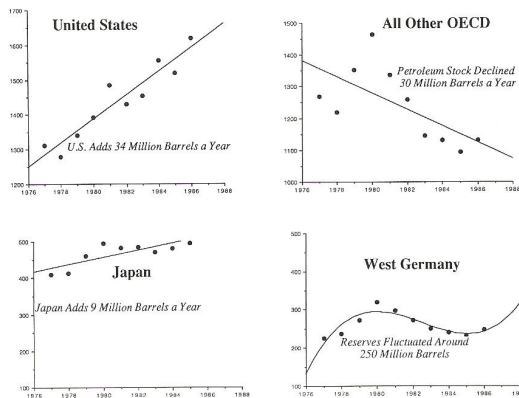
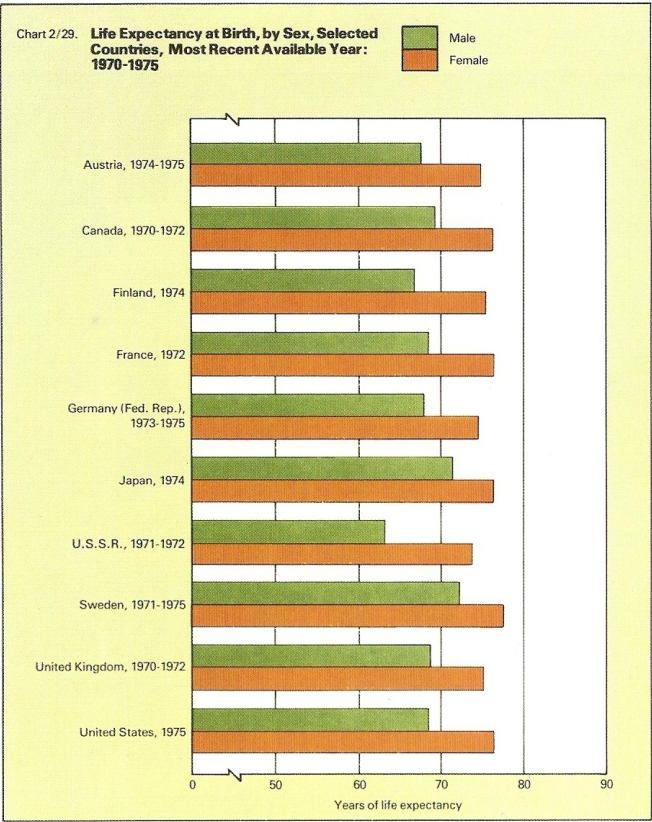


FIGURE 34. Regraphing of the data from figure 33 in which each country's data are shown relative to a straight line.

Figure 30: Wainer (1997), p. 36, Figure 34: Wainer (1997), p. 35, Figure 33, improved.

Rule 9: Alabama first!



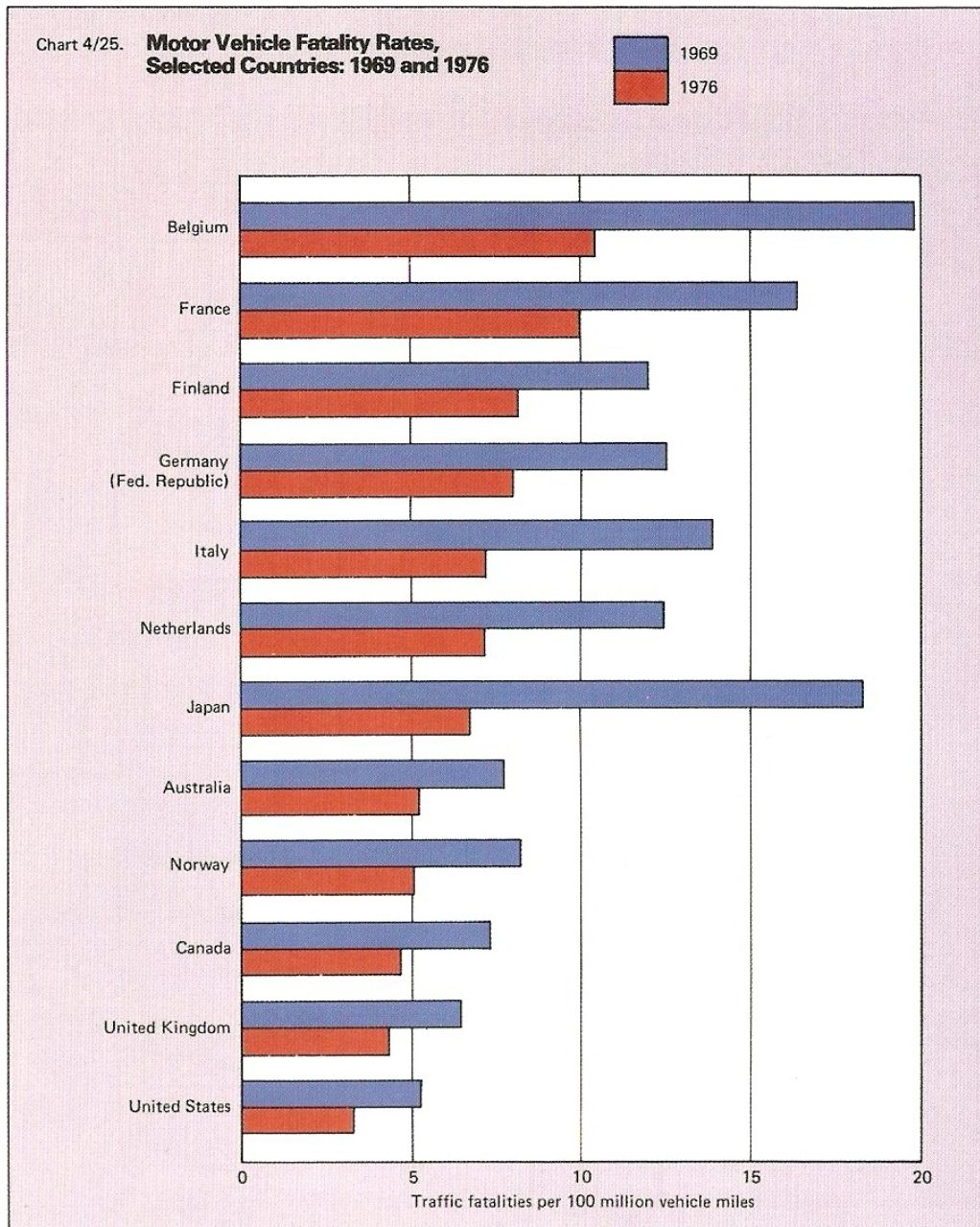
CHAPTER 1, FIGURE 35. Austria first! Obscuring the data structure in some life expectancy data by alphabetizing the plot.

Figure 31: Wainer (1997), p. 20B, Figure 35.



FIGURE 36. Ordering and spacing the data from figure 35 as a stem-and-leaf diagram provides insights previously invisible.

Figure 32: Wainer (1997), p. 37, Figure 36: Wainer (1997), p. 20B, Figure 35, improved.



CHAPTER 1, FIGURE 37. Ordering the bar chart by the data tells the tale a bit more clearly.

Figure 33: Wainer (1997), p. 20B, Figure 37: Layout similar to Wainer (1997), p. 20B, Figure 35, but improved due to ordering.

Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously.

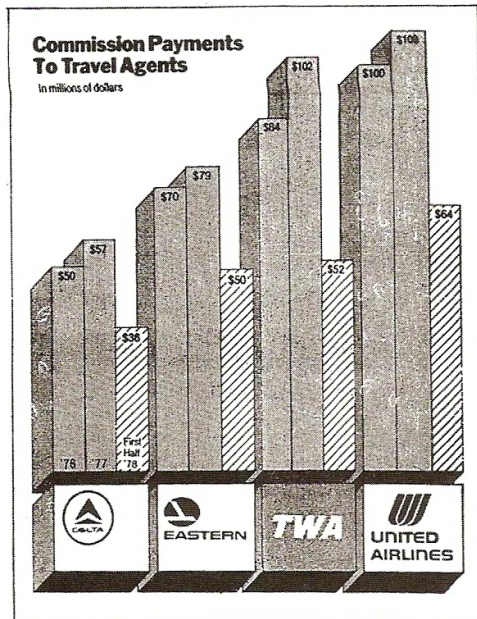


FIGURE 38. Mixing a changed metaphor with a tiny label reverses the meaning of the data.

Figure 34: Wainer (1997), p. 39, Figure 38.

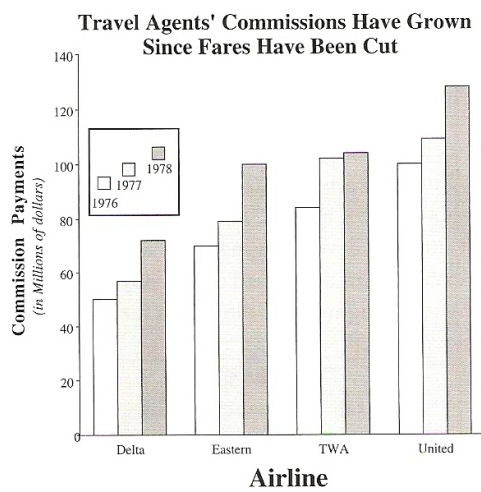


FIGURE 39. Figure 38 redrawn with 1978 data placed on a comparable basis shows that the fare cuts have been a boon to travel agents.

Figure 35: Wainer (1997), p. 39, Figure 39: Wainer (1997), p. 39, Figure 38, improved.

Rule 11: More is murkier: (a) more decimal places and (b) more dimensions.

TABLE 1		
Life Expectancy at Birth		
Country	Male	Female
Argentina	56.90	61.40
Brazil	39.30	45.50
Canada	67.61	72.92
Iceland	66.10	70.30
Japan	65.37	70.26
Mexico	37.92	39.79
Netherlands	71.40	74.80
New Zealand	68.20	73.00
Norway	71.11	74.70
Spain	58.76	63.50

Figure 36: Wainer (1997), p. 40, Table 1.

TABLE 2		
Life Expectancy at Birth		
Country	Male	Female
Netherlands	71	75
Norway	71	75
New Zealand	68	73
Canada	68	73
Iceland	66	70
Japan	65	70
Spain	59	64
Argentina	57	61
Brazil	39	46
Mexico	38	40

Figure 37: Wainer (1997), p. 40, Table 2: Wainer (1997), p. 40, Table 1, improved.

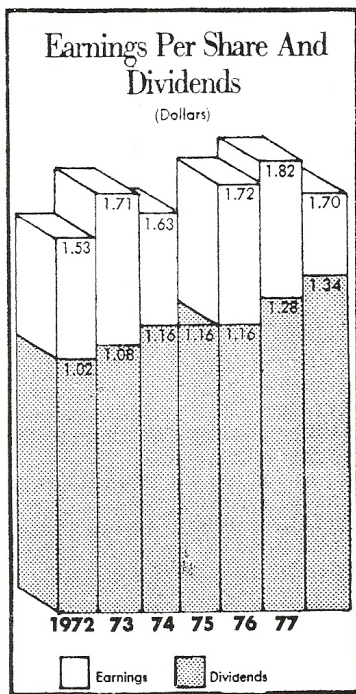


FIGURE 41. An extra dimension on earnings and dividends confuses even the grapher (from the *Washington Post*, 1979).

Figure 38: Wainer (1997), p. 41, Figure 41.

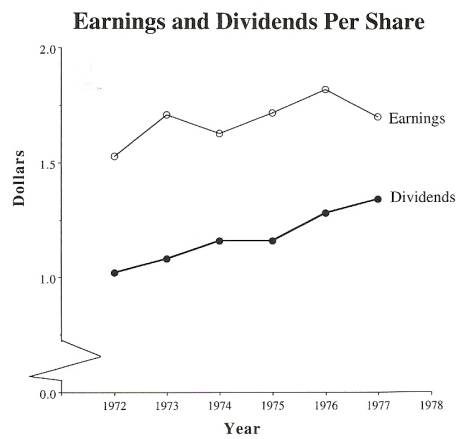
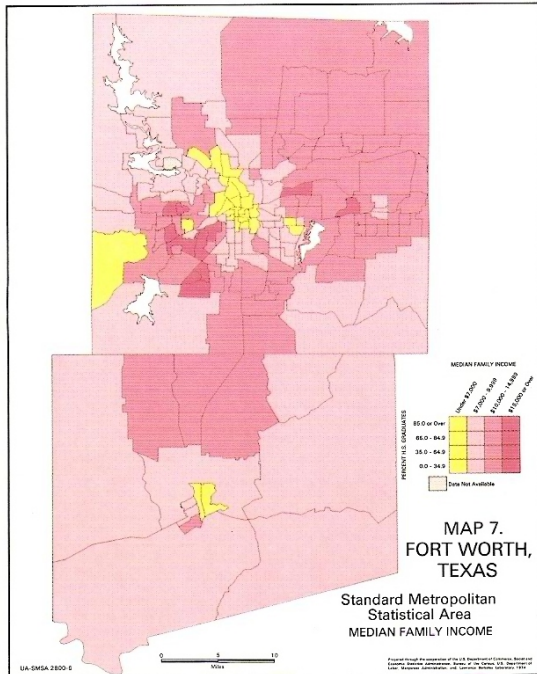


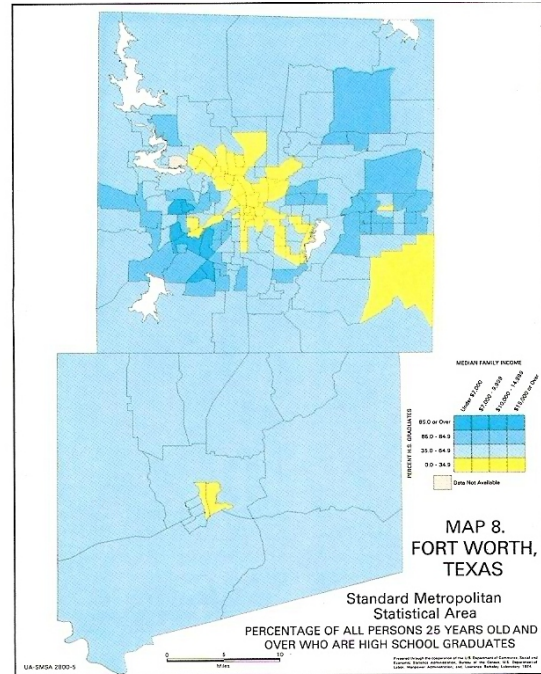
FIGURE 42. Data from figure 41 redrawn simply.

Figure 39: Wainer (1997), p. 42, Figure 42: Wainer (1997), p. 41, Figure 41, improved.

Rule 12: If it has been done well in the past, think of a new way to do it.

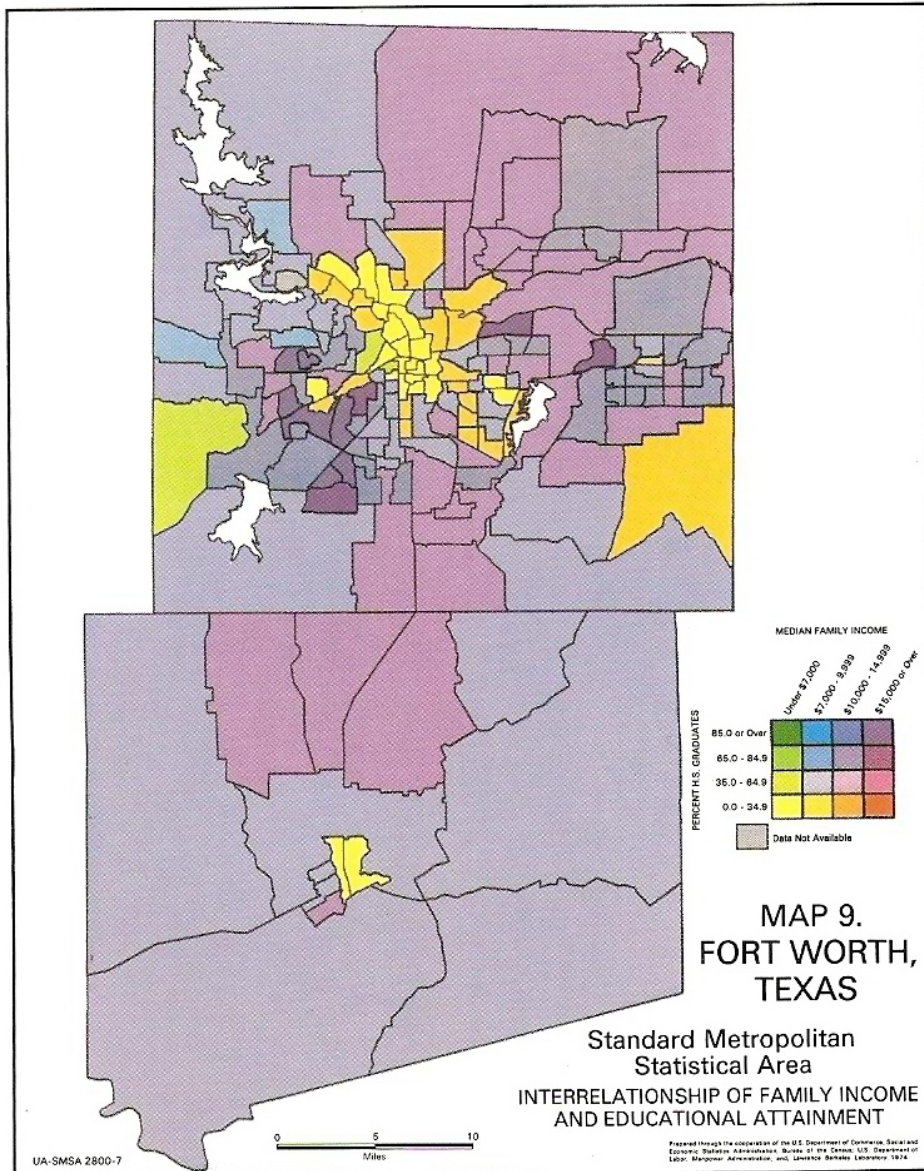


CHAPTER 1, FIGURE 44. The geographic distribution of median family income in Fort Worth, Texas, in 1974.



CHAPTER 1, FIGURE 45. The geographic distribution of percentage of high-school graduates in Fort Worth, Texas, in 1974.

Figure 40: Wainer (1997), p. 20C, Figures 44 & 45: Traditionl maps.



CHAPTER 1, FIGURE 46. The geographic distribution of both median family income and percentage of high-school graduates in Fort Worth, Texas, in 1974, shown as a two-variable color map.

Figure 41: Wainer (1997), p. 20C, Figure 46: Wainer (1997), p. 20C, Figures 44 & 45, modified but **not** improved.

1.4 Bad Graphics are Everywhere — In Space and in Time

Example 1: Zion National Park, UT, Shuttle Parking Lot, December 28, 2002

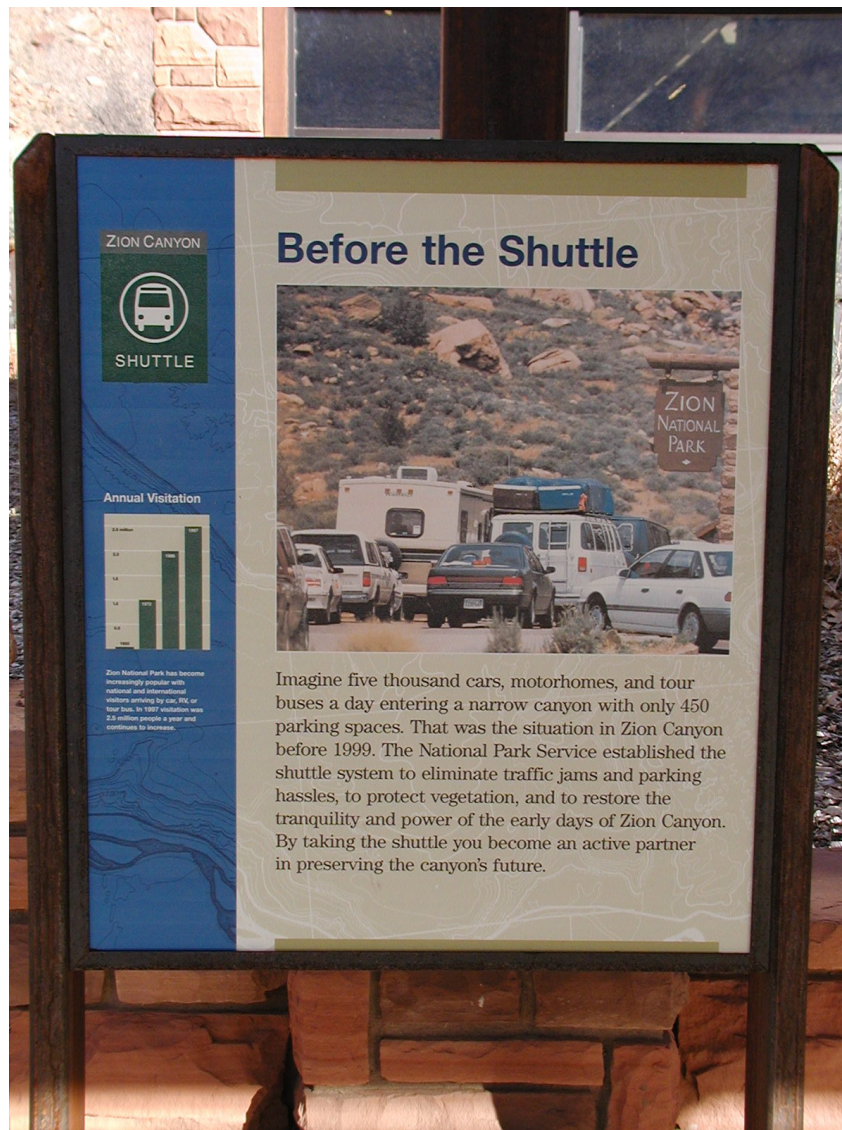


Figure 42: Personal Photograph: From the distance, the annual visitation appears to increase linearly, . . .

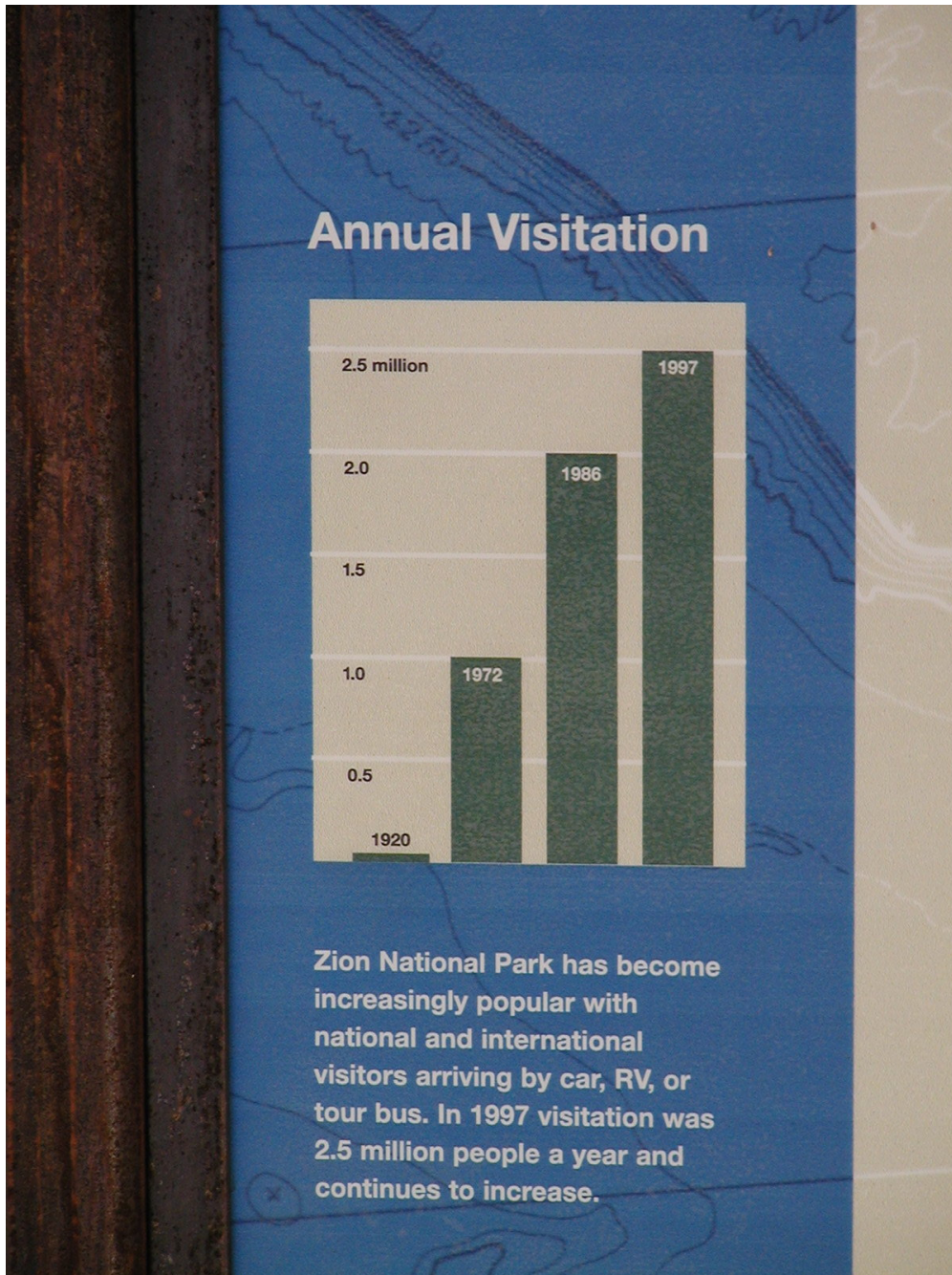


Figure 43: Personal Photograph: ... but at a closer view, this is certainly not the case.

Rules followed (to make this a bad graphic):

Improved Version:

Example 2: Berlin, Germany, August 20, 2006



Figure 44: Personal Photograph: Exhibit at the 1936 Berlin Olympic Site, related to the history of the Olympic area from 1909 to 1936 to 2006.



Figure 45: Personal Photograph: Historical graphic (from the late 1920ies), dedicated to the development of women's gymnastics as part of the *Deutsche Turnerschaft* (the governing body of German gymnastics).

Rules followed (to make this a bad graphic):

Improved Version:

Example 3: Wikipedia, 2009

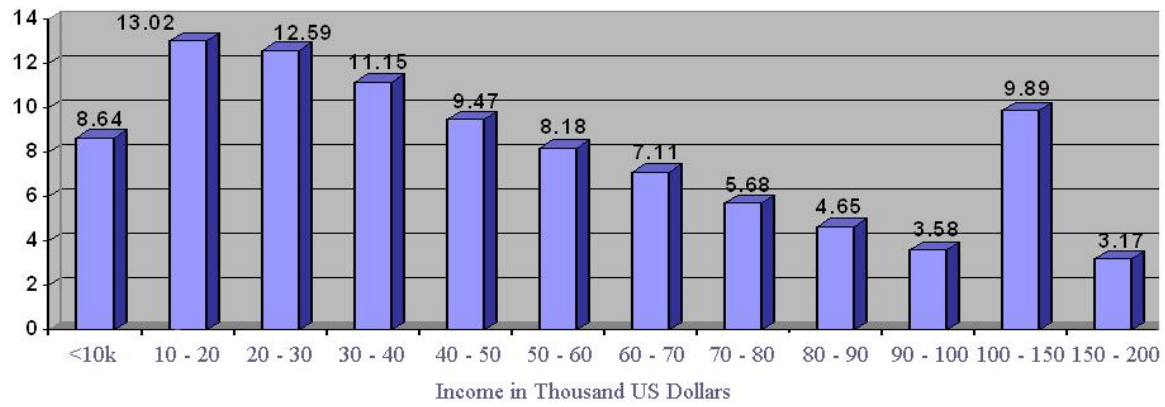


Figure 46: Figure taken from http://en.wikipedia.org/wiki/Household_income_in_the_United_States on 1/13/2009.

Rules followed (to make this a bad graphic):

Improved Version:

Example 4: Computational Statistics, 2002

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age, surface and contour graphs. Various types of graphs created by KyPlot are shown in Figures 3 and 4.

Almost every component of each graph can be customized through dialog boxes. Double-clicking an axis of a graph brings up a dialog box through which one can change various settings for the axis interactively. The scales of the x- and y-axes of graphs can be individually set as either linear or logarithmic. Error bars can be attached to either x- or y-values, or both, and the attributes of individual error bars can also be customized. (For example, in Figure 3A, the error bars for two data points of a line graph have been partially suppressed to avoid overlapping.) A break along an axis can be set, over a specific range and at a specific location, to indicate that a range of values has been omitted (Figure 3B).

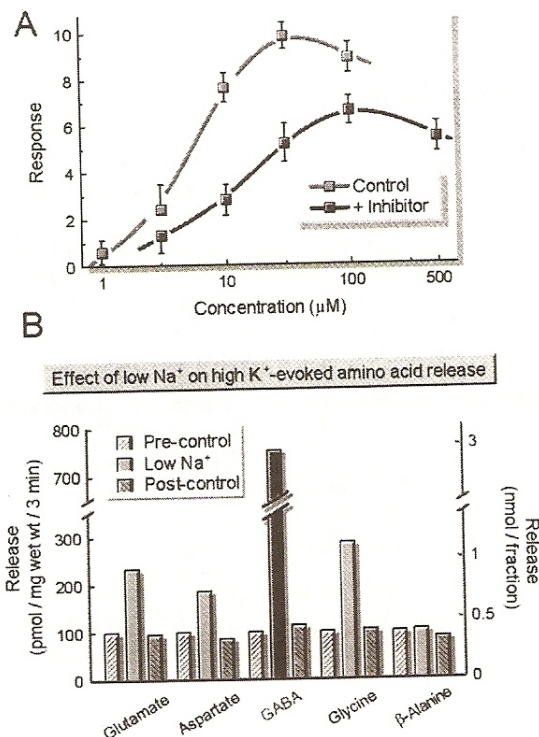


Figure 3: Line and bar graphs created with KyPlot

Figure 47: Yoshioka (2002), p. 430, Figure 3: Intended (!) features of the KyPlot software package for statistical data analysis and visualization.

Rules followed (to make this a bad graphic):

Improved Version:

1.5 Rules for Good Data Displays

Wainer (1997), p. 46, suggests:

- “1. Examine the data carefully enough to know what they have to say, and then let them say it with a minimum of adornment.
2. In depicting scale, follow practices of “reasonable regularity.”
3. Label clearly and fully.”

Tufte (1983), p. 77, suggests:

“Graphical integrity is more likely to result if these six principles are followed:

- The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the numerical quantities represented.
- Clear, detailed, and thorough labeling should be used to defeat graphical distortion and ambiguity. Write out explanations of the data on the graphic itself. Label important events in the data.
- Show data variation, not design variation.
- In time-series displays of money, deflated and standardized units of monetary measurements are nearly always better than nominal units.
- The number of information-carrying (variable) dimensions depicted should not exceed the number of dimensions in the data.
- Graphics must not quote data out of context.”

Robbins (2005), pp. 375–377, provides a “*Checklist of Possible Graph Defects*” in her Appendix A:

“Can the reader clearly see the graphical elements?”

- Do the data stand out? Are there superfluous elements?
- Are all graphical elements visually prominent?
- Are overlapping plotting symbols visually distinguishable?
- Can superposed data sets be readily visually assembled?
- Is the interior of the scale–line rectangle cluttered?
- Do data labels interfere with the quantitative data or clutter the graph?
- Is the data rectangle within the scale–line rectangle?
- Do tick marks interfere with the data?
- Do tick mark labels interfere with the data?
- Are axis labels legible?
- Are there too many tick marks?

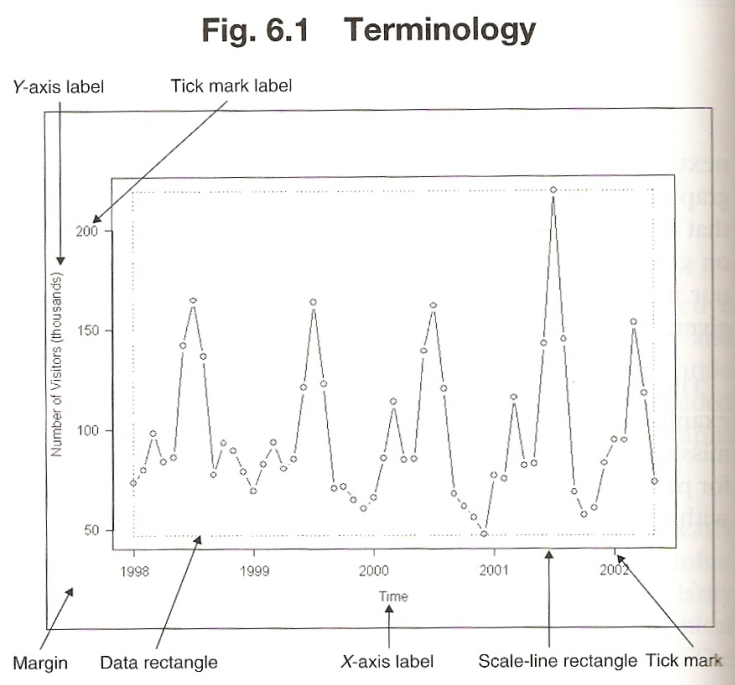


Figure 48: Robbins (2005), p. 156, Figure 6.1.

- Are there too many tick mark labels?
- Do the grid lines interfere with the data?
- Are there notes or keys inside the scale-line rectangle?
- Will visual clarity be preserved under reduction and reproduction?

Can the reader clearly understand the graph?

- Are the data drawn to scale?
- Is there an informative title?
- Is area or volume used to show changes in one dimension?
- Are there too many dimensions in the graph (more than in the data)?
- Are common baselines used wherever possible?
- Are all labels associated with the correct graphical elements?
- Is the reader required to make calculations?
- Are groups of charts drawn consistently?

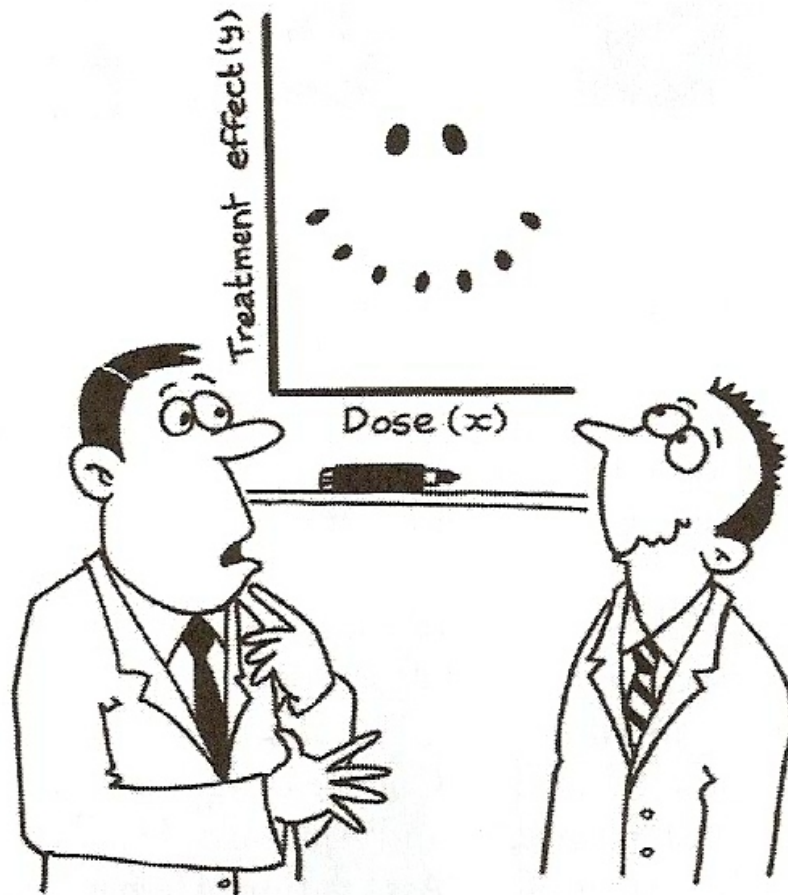
Are the scales well chosen and labeled?

- Is zero included for all bar graphs?
- Are there any unnecessary scale breaks?
- Is there a forceful indication of a scale break?
- Are there numerical values on two sides of a scale break that are connected?
- Does the aspect ratio allow the reader to see variations in the data?
- Are scales included for all axes?
- Are the scales labeled?
- Are tick marks at sensible values?
- Do the axes increase in the conventional direction?
- Does the data rectangle fill as much of the scale–line rectangle as possible?
- Are uneven time intervals handled correctly?
- Are the scales appropriate when different panels are compared?”

1.6 Further Reading

In addition to Wainer (1997), Tufte (1983), and Robbins (2005), cited so far in this chapter, many other sources exist that compare bad graphics with good graphics. Some of these additional sources are:

- Bertin (1977) and Bertin (2005) (first published in 1967)
- Henry (1995)
- Holmes (1991): check the author credentials and then decide whether this book is a source for good or bad graphics
- Huff & Geis (1954)
- Jones (2000)
- Kosslyn (1994) and Kosslyn (2006)
- Krämer (1991)
- Wainer (2005)
- Wainer (2007)
- Wallgren et al. (1996)
- Zelazny (2001)



"It's a non-linear pattern with outliers.....but for some reason I'm very happy with the data."

A CAUSE-commissioned cartoon that is part of the CAUSEweb collection and available for free noncommercial use by statistics teachers. Cartoon by John Landers ©. Provided by permission.

Figure 49: Amstat News, January 2009, p. 25, Cartoon.

Appendix

Homework Assignments

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- Yoshioka, K. (2002), ‘KyPlot — A User-Oriented Tool for Statistical Data Analysis and Visualization’, *Computational Statistics* **17**(3), 425–437.
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— THE END —

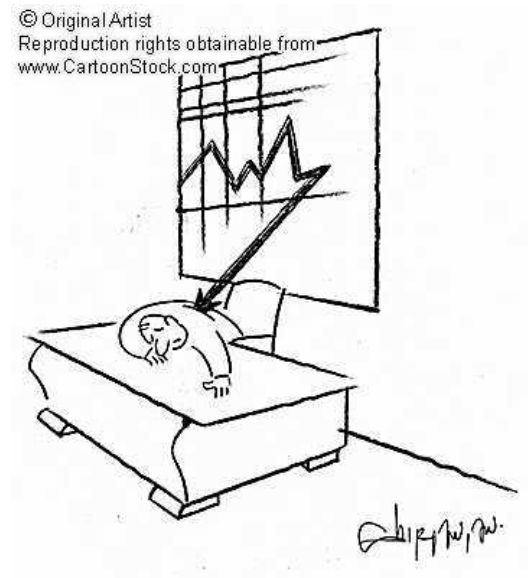


Figure 50: http://www.cartoonstock.com/blowup_stock.asp?imageref=vsh0184&artist=Shirvanian,+Vahan&topic=statistics+, Cartoon.